

Adapted for high school

Could Plastic Pollution Be Making Climate Change Worse?

Introduction

Plastic is one of the most useful materials people have created. It's everywhere, from phones and medical tools to food packaging and clothes. But it also causes serious problems.

Plastics do damage to the climate with greenhouse gases emitted during production, interference with how Earth stores carbon, and affecting how much sunlight Earth reflects or absorbs. A study conducted by Xia Zhu, Justin Konik, and Holly Kaufman explored how plastics impacted the planet in three ways: greenhouse gas emission, carbon storage, and radiation.

They reviewed 143 articles on this topic and summarized the results. According to their findings, each year plastics emit up to two gigatonnes of carbon dioxide equivalents (about 5% of the total CO₂ released) with most emissions produced at the primary production and product manufacturing stages. These findings illustrate that immediate action is required to remove carbon emissions from the life cycle of the plastics we use every day.

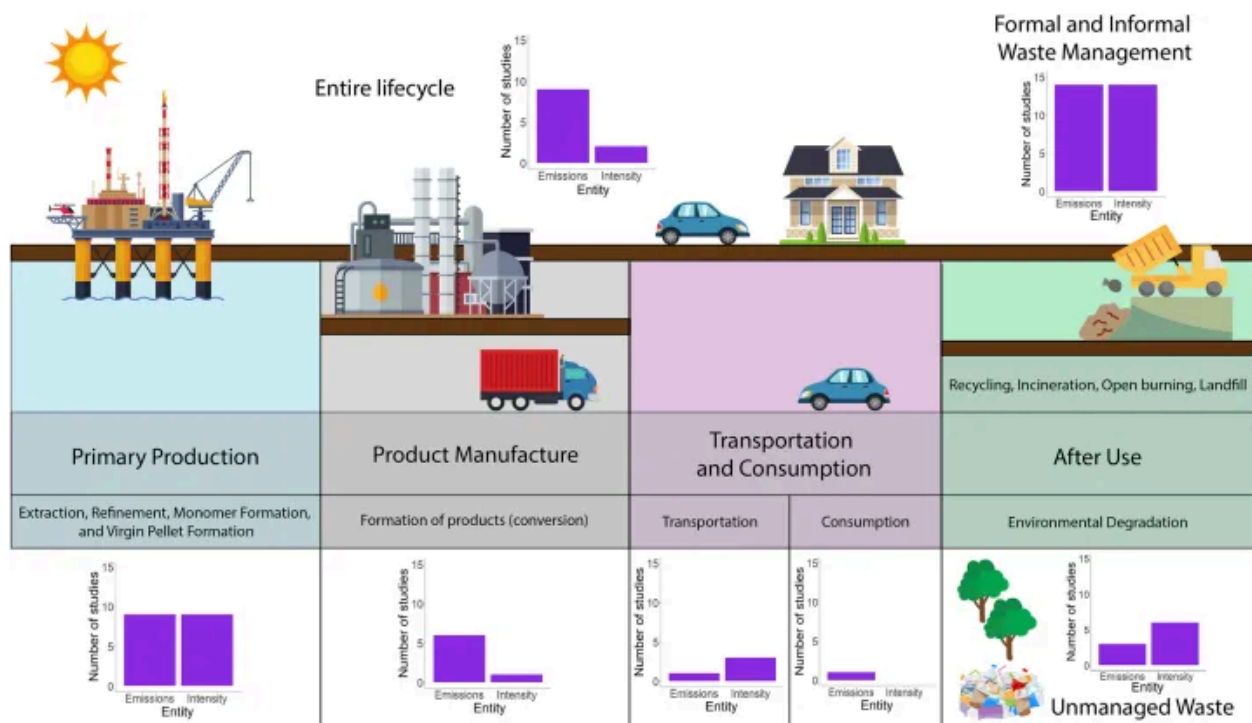
Materials and Methods

In this study, researchers conducted a systematic review, looking at studies that had already been published to summarize the results. They specifically examined peer-reviewed studies in English that were directly related to plastics and the climate.

After filtering everything, they ended up with 143 studies:

- 36 about greenhouse gas emissions
- 83 about carbon sinks
- 24 about the radiation budget

They then looked through these studies to figure out what scientists knew for sure and what still needed more research.



Results

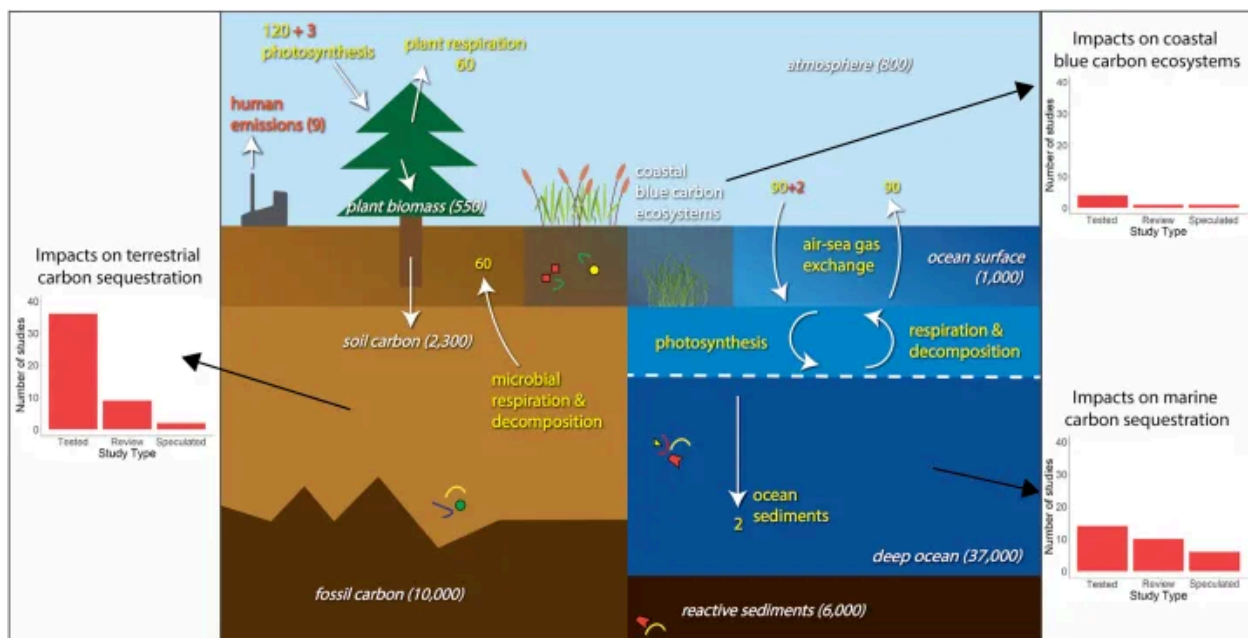
Lifecycle stage		Primary production	Product manufacture	Transportation	Consumption	Waste management	Unmanaged waste
Total emissions (MT CO ₂ e)	Global	1,085–1700	535	NA	NA	16–322	NA
	National	11.24–377.36	3.8–185	NA	NA	0.04–55.62	<0.01–0.002
Emissions intensities (kg CO ₂ e/kg plastic)		1.82–12.9	2.203	0.0331–0.128	NA	–0.04–6.13	0.112

Waste management option		Recycling (unspecified)	Mechanical recycling	Chemical recycling	Incineration	Open burning	Open dumping	Landfill
Total emissions (MT CO ₂ e)	Global	49	NA	30	96–322	NA	NA	16–17
	National	0.05–55	0.71–9	0.3	3.6–55.62	0.21–0.50	0.58	0.04–1.3
Emissions intensities (kg CO ₂ e/kg plastic)		0.003–0.63	0.005–0.91	–0.04–1.87	0.15–3.01	NA	1.515	0.015–1.16

Greenhouse Gas Emissions

The researchers found that the life cycle plastic releases greenhouse gases at every stage, from extracting the oil used and ending with disposal. The largest source of emissions comes from making raw plastic materials, producing between 1,085 and 1,700 million tons of CO₂ equivalents each year (equivalent to the emissions of 100 million Americans). The equivalents include methane and nitrous oxide. Manufacturing specific plastic products is the second biggest source. Plastic disposal also has an effect, with burning plastic releasing the most greenhouse gases, landfilling releasing the least, and recycling falling somewhere in between.

How Plastics Affect Carbon Storage



Earth naturally stores carbon in places like soil and oceans. These are called carbon sinks. If plastics damage these systems, more carbon dioxide stays in the atmosphere, which worsens climate change.

In soil, plastics affect bacteria and increase their CO₂ release in many cases. These bacteria are supposed to consume carbon, storing it in soil

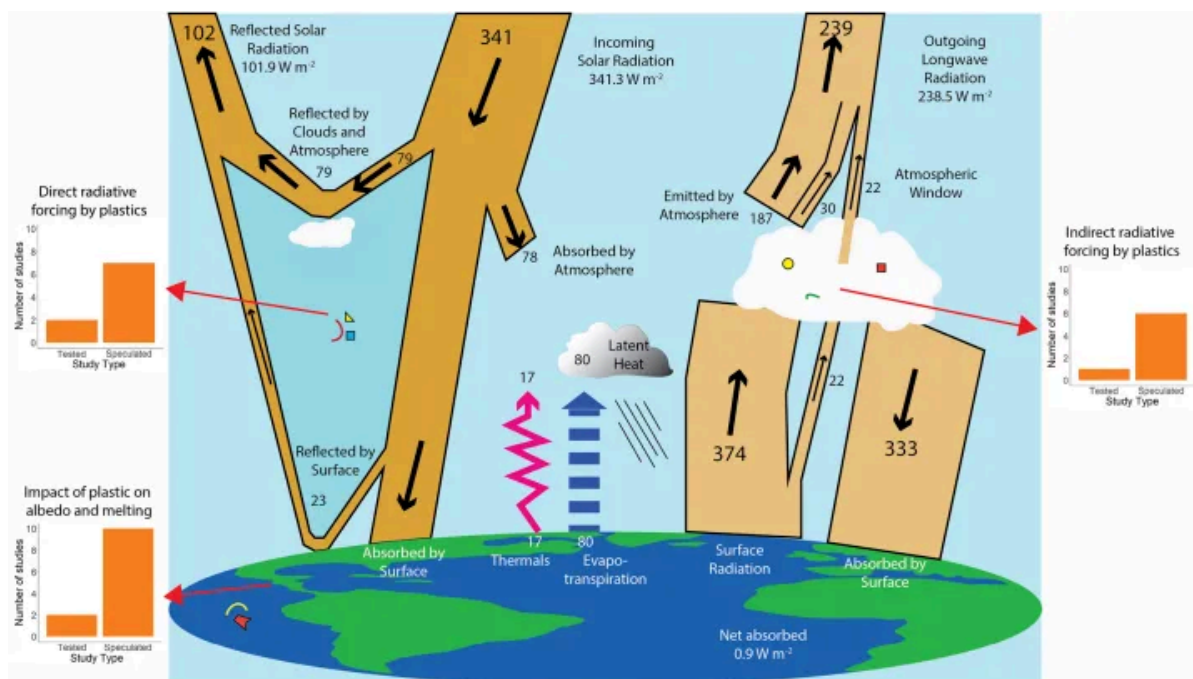
and oceans where they live. However, microplastics disrupt this process, causing less CO₂ to be stored. Marine effects are even worse. Plastics disrupt the biological carbon pump—where phytoplankton absorb carbon dioxide and bring it to deeper areas of the ocean—in every study that tested it, due to harming the phytoplankton.

Overall, there were 198 cases where plastics made climate change worse through carbon cycling and 94 where they helped with the cycle.

How Plastics Affect the Radiation Budget

This is the newest and least understood area. Most of the 24 studies about this were based on predictions rather than looking at real results.

The radiation budget is the balance between sunlight entering Earth's atmosphere and heat leaving it. So far, it seems that small plastic particles in the air could slightly reduce the amount of sunlight reaching Earth, which could have a small cooling effect. When plastic gets into snow or ice, it may make surfaces more reflective, which also slightly cools the planet. However, these effects are small and very uncertain, especially compared to the warming caused by greenhouse gases.



Discussion

The main takeaway is that plastics accelerate climate change through greenhouse gas emissions and by disrupting carbon sinks.

The biggest impact comes from the early stages of plastic production. Making plastic and manufacturing products causes many emissions, making the most effective solution to reduce how much plastic we produce and use, especially new plastic.

When it comes to disposal, there are few good solutions, with burning plastic being worse, and landfill not much better. Recycling produces fewer greenhouse gas emissions and can reduce the need to make new plastic. However, mechanical recycling only delays disposal rather than preventing it. Recycling can help, but reducing how much plastic we produce remains the most effective action.

Plastic and climate change are closely connected. Plastics release up to 2 billion tons of greenhouse gases each year, and harm natural carbon storage systems. The most important steps to take are to reduce plastic production and use, choose better disposal methods, and include plastics in climate policies.

Glossary

Biological carbon pump — The ocean's process of moving carbon from the surface to the deep sea, where it can be stored for centuries.

Carbon sink — A natural system that absorbs CO₂ from the atmosphere into forests, soils, and oceans.

CO₂e (Carbon Dioxide Equivalent) — A standard unit of measurement used to express the climate impact of all greenhouse gases in a single metric based on how much they contribute to warming.

Greenhouse gas — A gas that traps heat in Earth's atmosphere (like Carbon dioxide and methane).

Landfill — A site where waste is disposed of.

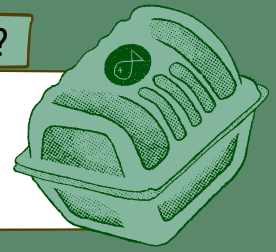
Life cycle — All the stages a product goes through: making raw materials, manufacturing, using, and disposing.

Phytoplankton — Tiny plant-like organisms in the ocean that absorb CO₂ and are the base of the ocean food chain.

Primary production — The first stage of making plastic, where fossil fuels are turned into raw plastic material.

Radiation budget — The balance between sunlight coming in and heat going out of Earth's atmosphere, which controls the planet's temperature.

Systematic review — A research method where scientists gather and analyze all existing studies on a topic to find overall patterns.



Reading Questions

1. What is a carbon sink? Why does it matter if plastics are damaging them?

2. The article says the best solution is to make and use less plastic. Is that realistic? What are some alternatives to single-use plastics already being used?

3. Scientists still don't fully understand how plastics affect clouds. If you could design a study to test this, what would you do?

4. Burning plastic releases the most greenhouse gases, but landfiling creates other problems. What factors should we consider when deciding how to dispose of plastic?